

ThreatWatch CyberLab

Software Design Document (SDD)



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1.) Problem Statement

Many cybersecurity students and entry-level analysts have a hard time gaining real-world experience since they don't have access to realistic security environments. Enterprise security systems like Security Information and Event Management (SIEM) platforms and real Security Operations Center (SOC) infrastructures are typically hard to use, cost a lot of money, and are not easy for beginners to get to. Because of this, students may know a lot about cybersecurity in theory but not have any real-world experience with things like assessing alerts, responding to situations, and figuring out how defensive decisions affect things.

ThreatWatch CyberLab fills this gap by offering a lightweight, simulation-based environment for intrusion detection that lets users practice finding and dealing with cyber threats in a safe setting. The tool lets students practice their investigative and defensive abilities in a safe way by mimicking real-life attack situations.

2.) User Personas

ThreatWatch CyberLab is designed for users with varying levels of cybersecurity knowledge who want to develop practical experience identifying and responding to cyber threats. The following personas represent typical users who would benefit from the system.

2.1) Persona 1 - Sekani Gray (Cybersecurity Intern)

Sekani Gray is a 23-year-old cybersecurity intern who recently earned a bachelor's degree in Cybersecurity. He has a strong foundation in security concepts such as networking, intrusion detection, and incident response, but limited real-world experience working inside a Security Operations Center (SOC).

As an intern, Sekani wants to build confidence by applying what he learned in school to realistic scenarios. He is motivated to improve his technical skills, better understand security alerts, and prepare for a full-time cybersecurity role.

Demographics

- **Age:** 23
- **Education:** Bachelor's degree in Cybersecurity
- **Job Role:** Cybersecurity Intern
- **Technical Skill Level:** Entry-level to intermediate

Goals

- Practice working with intrusion detection systems
- Learn how to interpret and prioritize security alerts

- Gain hands-on experience responding to cyber threats
- Build confidence before transitioning into a full-time role

Frustrations / Pain Points

- Most training is theoretical and not hands-on
- Real SOC tools can be overwhelming for beginners
- Limited opportunities to safely practice making mistakes
- Difficulty connecting classroom knowledge to real incidents

How Sekani Uses ThreatWatch CyberLab

Sekani utilizes ThreatWatch CyberLab as a hands-on training environment, where he can observe simulated cyberattacks and observe how an intrusion detection system responds. He reviews alerts, decides how to respond, and learns from feedback on his actions. This allows Sekani to practice real-world cybersecurity workflows without the risk of impacting a live system.

Why This Persona Matters

This persona represents an early-career cybersecurity professional who needs practical experience to bridge the gap between education and real-world work. Designing ThreatWatch CyberLab with Sekani in mind ensures the platform focuses on usability, learning, and realistic security workflows rather than unnecessary complexity.

2.2) Persona 2 - Christiana Love (Undergrad Student)

Christiana Love is a 20-year-old undergraduate Computer Science student who is interested in cybersecurity but has limited exposure to security concepts and tools. She has completed introductory programming and networking coursework but has not yet worked with intrusion detection systems, security logs, or incident response processes.

Christiana is curious about cybersecurity as a potential career path, but feels overwhelmed by the complexity of professional security tools. She wants a supportive, low-risk environment where she can learn how cyberattacks are detected and handled without needing advanced prior knowledge.

Demographics

- **Age:** 20
- **Education:** Undergraduate Computer Science major
- **Job Role:** Student (pre-internship)
- **Technical Skill Level:** Beginner

Goals

- Understand how cybersecurity works in real-world settings
- Learn basic intrusion detection and response concepts
- Gain hands-on experience without fear of breaking systems
- Build confidence before taking advanced security courses

Frustrations / Pain Points

- Security terminology and tools feel intimidating
- Learning resources often assume prior security knowledge
- Difficulty visualizing how attacks and defenses interact
- Fear of making mistakes while learning

How Christiana Uses ThreatWatch CyberLab

Christiana uses ThreatWatch CyberLab as an introductory training platform. She begins with guided scenarios that explain alerts in simple language and offer clear response options. Through repeated exposure and feedback, Christiana learns to recognize common attack patterns and understand why certain responses are effective, helping her gradually build foundational cybersecurity skills.

Why This Persona Matters

Christiana represents beginner users who are exploring cybersecurity for the first time. Designing ThreatWatch CyberLab with Christiana in mind ensures the platform remains accessible, educational, and supportive, making cybersecurity concepts easier to understand for users without a strong technical background.

2.3) Persona 3 - Dr. Angel Ryder (Cybersecurity Instructor)

Dr. Angel Ryder is a cybersecurity instructor and professor who teaches undergraduate and early graduate-level security courses. She has a strong background in cybersecurity concepts and professional experience, but is primarily focused on helping students translate theory into real-world practice.

Dr. Ryder wants a safe, structured platform that allows students to experience intrusion detection and response workflows without the cost, complexity, or risk of enterprise security systems.

Demographics

- **Age:** 47
- **Education:** Master's in Cybersecurity
- **Job Role:** Cybersecurity Instructor / Trainer
- **Technical Skill Level:** Advanced

Goals

- Provide students with hands-on cybersecurity experience
- Demonstrate how intrusion detection systems work in practice
- Teach decision-making and response workflows, not just theory
- Evaluate student understanding through realistic scenarios

Frustrations / Pain Points

- Limited access to realistic SOC-style environments for teaching
- Enterprise tools are too complex or expensive for classroom use
- Traditional labs focus on exploitation rather than defense
- Difficulty assessing how well students understand response processes

How Dr. Angel Ryder Uses ThreatWatch CyberLab

Dr. Ryder uses ThreatWatch CyberLab as an instructional and training tool. She assigns scenarios to students, allows them to interact with simulated intrusions, and reviews performance

summaries to assess how students interpret alerts and respond to incidents. The platform supports guided learning while allowing Dr. Ryder to scale hands-on practice across multiple students.

Why This Persona Matters

Dr. Ryder represents instructors and trainers who shape how cybersecurity is taught. Designing ThreatWatch CyberLab with her needs in mind ensures the platform supports structured learning, guided progression, and assessment, making it suitable for classroom and training environments.

3.) Functional Requirements

The following functional requirements outline the primary functionalities and behaviors of the ThreatWatch CyberLab system. These requirements explain how users will use the platform and how the system will operate in simulated cybersecurity situations.

- **FR-01: Training Profile Selection**

- Before a training session begins, the system shall require the user to select a training profile. The system shall configure the simulation's guidance level and difficulty settings based on the selected profile.

- **FR-02: Intrusion Simulation and Detection**

- When a training session is active, the system shall generate simulated intrusion events and classify each event as one of three supported intrusion types.

- **FR-03: Randomized Attack Triggering**

- During an active session, the system shall trigger simulated intrusion events at randomized time intervals.

- **FR-04: Alert Notification**

- When a simulated intrusion event occurs, the system shall display an alert notification to the user.

- **FR-05: Investigation Workflow**

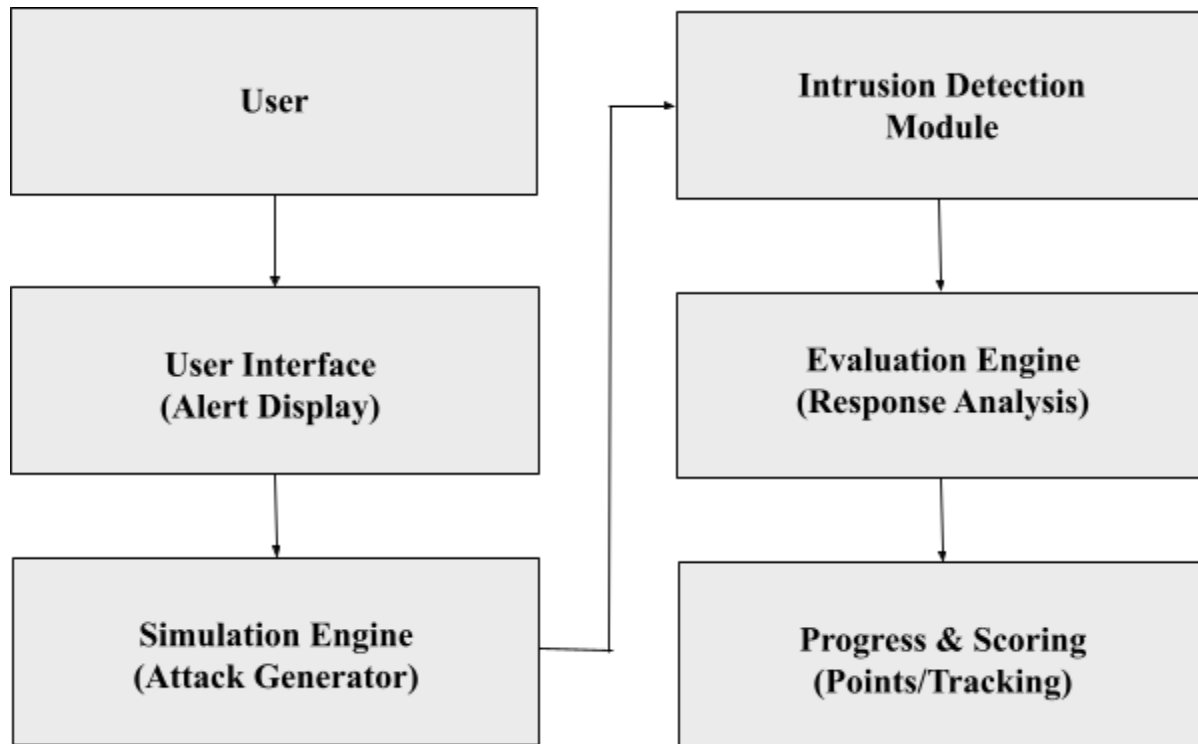
- When an alert is displayed, the system shall allow the user to initiate an investigation and view relevant intrusion details, including indicators associated with the simulated attack.

- **FR-06: Response Selection**

- After reviewing the intrusion details, the system shall allow the user to select one response option from multiple available choices.
- **FR-07: Response Evaluation**
 - When a response is selected, the system shall evaluate the user's decision and determine whether the response is correct or incorrect.
- **FR-08: Escalation Behavior**
 - If the user selects an incorrect response, the system shall increase the severity level of the intrusion event and simulate further attack progression.
- **FR-09: Outcome Display**
 - After evaluating the selected response, the system shall display the resulting state of the intrusion event.
- **FR-10: Explanation Feedback**
 - After displaying the outcome, the system shall provide an explanation describing why the selected response succeeded or failed.
- **FR-11: Points Reward**
 - If the user selects a correct response, the system shall award points to the user.
- **FR-12: Progress Tracking**
 - During a training session, the system shall record user responses and update the user's progress metrics accordingly.
- **FR-13: Streak Tracking**
 - If consecutive correct responses occur, the system shall update and display the user's current response streak.
- **FR-14: Alert Dismissal**

- If the user selects a dismissal option, the system shall remove the alert from the active alert list and record the action in the session log.

4.) System Architectural View



ThreatWatch CyberLab was constructed in a modular way, with many parts that work together. The user interface shows notifications and lets users look into and respond to fake attacks. This is how the user interacts with the system. The simulation engine makes up cyberattack scenarios that the intrusion detection module then works on. The system sends notifications to the user when it sees a threat. The user then has to look at the situation and decide how to respond. The evaluation engine looks at the user's answer and decides if the action worked. Finally, the progress and scoring module keeps track of how well users are doing, gives points for right answers, and keeps track of how far they've come in their training.

4.1) Intrusion Detection Module

The Intrusion Detection Module is responsible for analyzing simulated attack events generated by the Simulation Engine and determining whether the activity represents a cybersecurity threat. This module acts as the detection layer of the ThreatWatch CyberLab platform.

Input

The module receives simulated attack event data from the Simulation Engine. Each event contains structured information describing the simulated network activity. The input may include the following attributes:

- Attack type
- Source IP address
- Target system or port
- Number of connection attempts
- Timestamp of the activity
- Event identifier

Processing Steps

When an attack event is received, the Intrusion Detection Module performs the following steps:

1. The module receives the simulated attack event from the Simulation Engine.
2. The module parses the event data and extracts key attributes such as attack type, source IP address, and connection attempts.
3. The module compares the event characteristics against predefined detection rules.

4. If the event matches a known intrusion pattern, the module classifies the event as a detected intrusion.
5. The module assigns a severity level (Low, Medium, High) based on the characteristics of the attack.

Output

After detection and classification, the module generates an intrusion alert that is sent to the Evaluation Engine. The output includes:

- Intrusion type
- Severity level
- Source IP address
- Event identifier
- Detection timestamp

The Evaluation Engine then uses this information to present the alert to the user and evaluate the user's response to the simulated attack.

5.) Acceptance Criteria

Example 1

User story:

As a cybersecurity trainee, I want the system to evaluate the response I select for an intrusion alert so that I can learn whether my decision was correct.

Acceptance criteria:

- The system should evaluate the user's selected response after it is submitted.
- The system should determine whether the selected response is correct or incorrect based on the predefined scenario rules.
- The system should display the evaluation result to the user.
- The system should proceed to display the resulting state of the intrusion event after the response evaluation is completed.

Example 2

User story:

As a cybersecurity trainee, I want the system to escalate an intrusion event when I select an incorrect response so that I can understand the consequences of ineffective security decisions.

Acceptance criteria:

- If the user selects an incorrect response, the system should increase the severity level of the intrusion event.

- The system should simulate additional attack activity to represent escalation of the intrusion.
- The system should generate updated alerts reflecting the increased severity of the attack.
- The system should allow the user to continue responding to the escalated intrusion event.